

Mechanisms underlying movement patterns of the South American tortoise *Chelonoidis chilensis*

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Abstract

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Introduction

Animal trajectories are the result of a complex interplay between internal and external factors, including the animal's physiological state, environmental conditions, and interactions with other individuals. Understanding the mechanisms underlying movement patterns is crucial for predicting animal behavior, managing wildlife populations, and conserving biodiversity. How animals take decisions about when and where to move is a fundamental question in ecology and behavioral biology. Movement patterns can be influenced by a variety of factors, including resource availability, predation risk, social interactions, and environmental conditions. In this study, we focus on the South American tortoise *Chelonoidis chilensis*, a species that is classified as endangered. As a reptile, it is ectothermic and relies on external sources of heat to regulate its body temperature. This makes it particularly sensitive to environmental conditions, which can influence its movement patterns and behavior. By performing computer simulations it is possible to explore the effects of different factors on movement patterns and to identify candidate mechanisms that drive these patterns. Previous studies showed that tortoises movement is subdiffusive which suggests....

By analyzing GPS tracking data collected over several campaigns, we aim to uncover the statistical properties of their movement patterns and identify potential factors influencing their behavior.

Methodology

The study was carried out in a site of approximately 25 hectares, located about 20 km north of the city of San Antonio Oeste, Province of Río Negro, Argentina, where there is a stable population of *C. chilensis* with a large number of specimens. The area in general belongs to the Monte Austral phytogeographic unit, characterized by a shrub steppe with a predominance of *Larrea* spp. with several strata, the lower one being grasses and herbs, all with very little coverage, and particularly with very few cacti (Oyarzabal et al. 2018). The specific study area is characterized by a vegetation cover, mostly with xerophytic characteristics, with a predominance of grass clumps (Poaceae) and shrubs (León et al. 1998, Morello et al. 2012). In turn, the grass stratum has a higher species richness than the other Zygothallaceae of the Monte steppes. The soil is rather sandy with a low percentage of clay and abundant pebbles. Close to the border of the study area, there is the Gran Bajo del Gualicho, a Patagonian plain of up to 70 meters below sea level, characterized by a high saline concentration. This is why there are more saline sectors, where we can find halophyte species such as *Suaeda divaricata* and several species of the genus *Atriplex* spp. Scarce precipitations give an average of approximately 255 mm in the year, with minor peaks in spring and fall. Annual average temperature is around 14.5 °C with significant daily thermal amplitudes and a persistence of the west and southwest winds, making the region more arid (Godagnone and Bran, 2008). The minimum average temperature is 1°C, and the maximum is 30 °C, with extreme recorded temperatures of -11.5 and 44.6 °C (data for 1961–2021 provided by the National Meteorological Service of Argentina). We tracked six tortoises continuously during a whole year using GPS data loggers (i-gotU GT-120 Mobile Action, Taiwan), recording geographical coordinates every 15 minutes. The positions were recorded continuously from 6am to 10pm and from 2am to 3am. The equipment was attached to the carapace of each individual using a camouflage tape (@Duck Tape, Real Tree Hardwood Camouflage) to mimic the environment, with the precaution of not leaving any sharp corners, in order to prevent the tortoise from getting trapped in the vegetation. After attaching the device, tortoises were released at the same place that were found. The weight of the equipment never exceeded

In this work we present analysis and results spanning a whole year, involving a total of 20149 hours of GPS data loggers recording. Those recorded geographical positions corresponded to 813 trajectories from 6 individuals. The GPS data loggers were charged every fifteen days except during the brumation period, when animals were not disturbed and were observed every two weeks to check if they remained brumating at the same location. All recorded trajectories were preprocessed to exclude those corresponding to speeds above a threshold of 15 m/min, assessed as a reasonable maximum speed for the species, according to the speed empirical distribution presented in Fig.S3.

Results and Discussion

The distance traveled as a function of time was calculated from the (latitude, longitude) data by determining the distance on the Earth's surface using the Haversine function.

The total accumulated distance traveled was then calculated as:

$$\text{distot}(t) = \text{distot}(t - 1) + \text{distance}(t)$$

Figure ?? shows all the measured quantities for turtle T10. There are some temporal gaps corresponding to missing data.

A similar behavior is observed for the other turtles (T6, T12, T30, T54, T79, T128, T185, and T206).

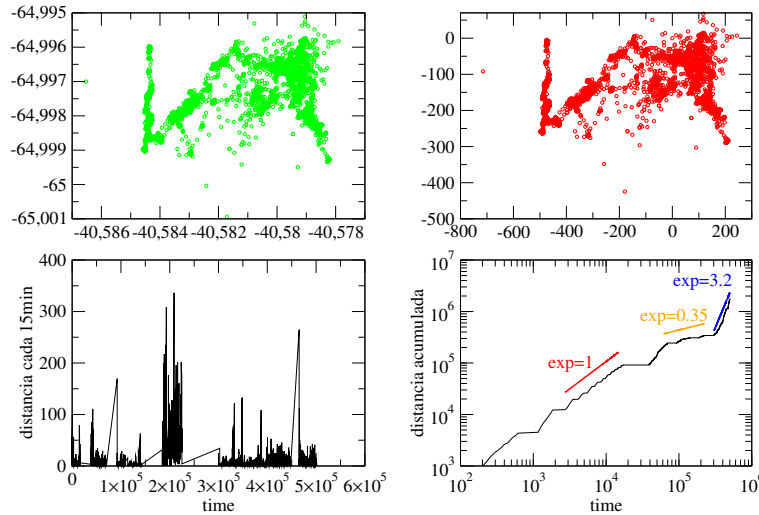


Figure 1: Datos de la Tortuga T10: A) Datos crudos de (lat,long) ; B) Lo mismo que en A) pero metros, para chequear que el pasaje de datos funcione. C) Distancia recorrida (cada 15min) en funcion del tiempo; D) Distancia acumulada vs time: los colores indican los exponentes del ajuste en cada periodo.

The distances traveled by the turtles were measured every 15 minutes. Based on the observed behavior, we can define an exponent as:

$$P(d) \approx d^{-\beta} \quad (1)$$

The mean exponent for the 9 turtles is: $\beta = 2.1(2)$, as shown in Figure ??.

La nueva figura, comparando los datos completos y solo hora solar (9 a 17hs) Figure ??.

Mean square deviation (MSD)

By calculating the MSD for each tortoise, markedly different behaviors are observed, both among tortoises and depending on the time period (Figure ??). We can fit an exponent:

$$\text{MSD} \approx t^\alpha \quad (2)$$

The data suggest that, at least during certain periods, the behavior is subdiffusive ($\alpha < 1$).

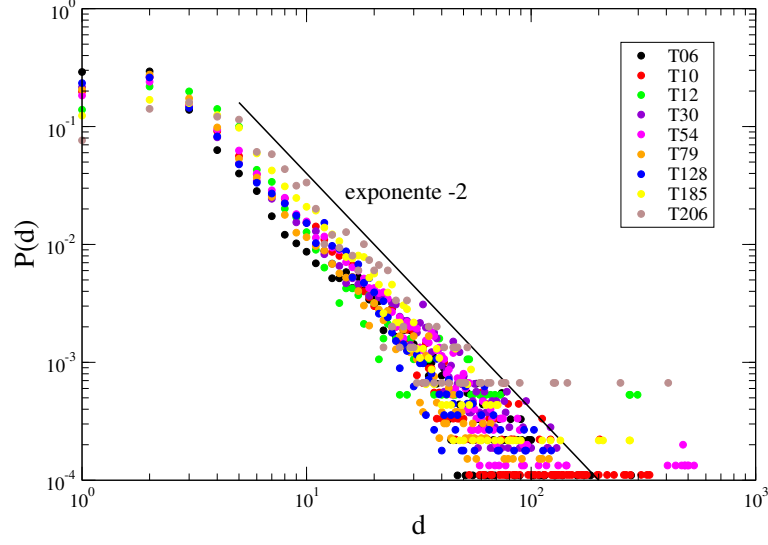


Figure 2: Distribucion de distancias recorridas cada $\Delta t = 15min$.

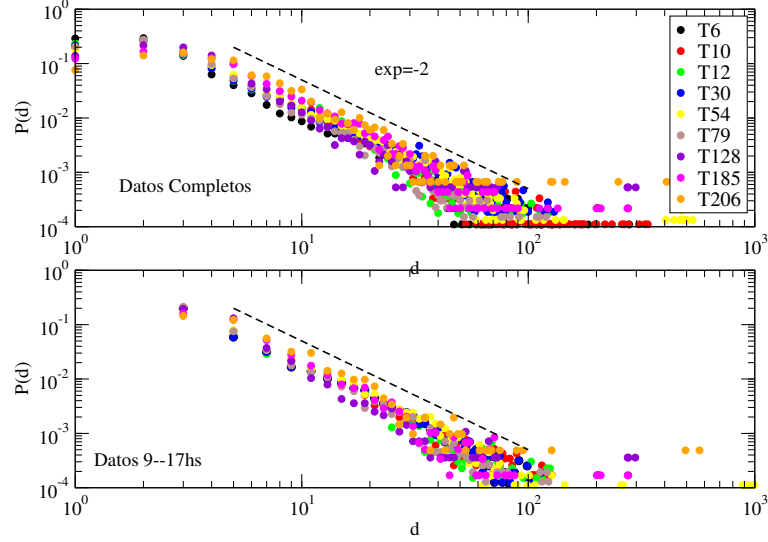


Figure 3: Distribucion de distancias recorridas cada $\Delta t = 15min$.

Seasonality

If we attempt to separate the behavior according to biological/climatic periods, namely: November–January, February–March, and April–October, we observe that the MSD behavior is markedly different, with exponents ranging from subdiffusive to diffusive regimes. Figure ?? shows only the case for T10. The seasonal periods still need to be refined. Here, only the calendar months were considered, but factors such as temperature, late spring conditions, and related environmental variables should probably also be taken into account.

Para todas las tortugas el MSD Figure ?? y la distribucion de distancias $P(d)$ Figure ??

In both figures, Fig. ?? and Fig. ??, the dotted lines correspond to the best fits obtained. Table 1 lists the values of these exponents ($MSD(t) \approx t^\alpha$ and $P(d) \approx d^{-\beta}$). As can be seen, there is considerable dispersion among turtles and across different periods of the year.

It should be noted that for some tortoises (T128, T185, and T206) only limited data are available. For example, no data are available for the November–January period, and for the April–October period only data from the first months (April and May) are available, as indicated in Figures ?? and ??.

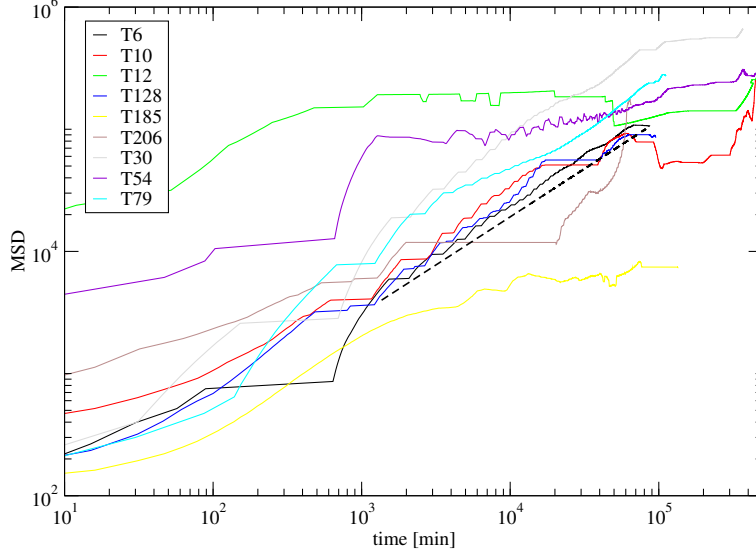


Figure 4: All turtles (identified as Tnumber): MSD versus time. The dotted black line has a slope corresponding to $\alpha \approx t^{0.8}$.

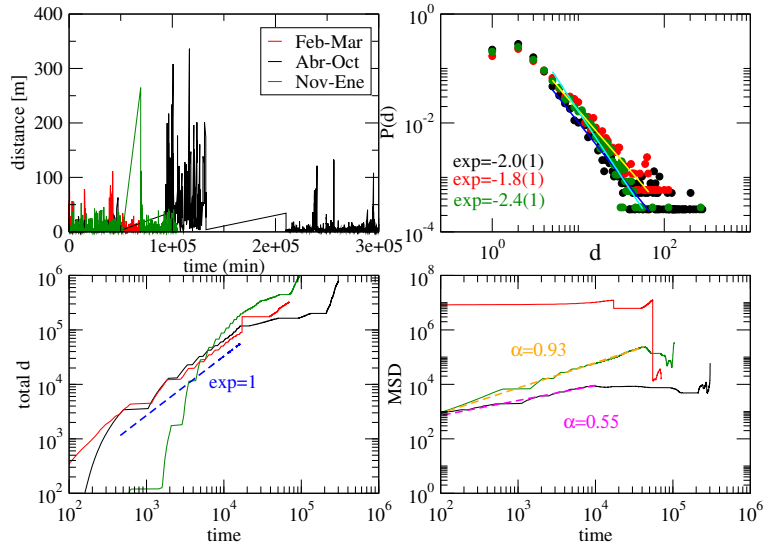


Figure 5: Separacion estacional de T10.

Males and females

So far, the MSD appears to exhibit a power-law behavior during the February–March period. Beyond this interval, it becomes difficult to obtain reliable fits.

Within this period, it may be useful to separate the data between males and females in order to investigate possible differences.

Figure ?? shows the MSD behavior separated by males and females. The exponent α (Eq. ??) does not appear to change significantly between males and females; rather, there is considerable dispersion in the data (see Table 2).

A similar situation is observed for the distribution of traveled distances (Figure ??), although in this case the dispersion in the curves may be larger among females than among males.

Table 2 lists the values of the exponents.

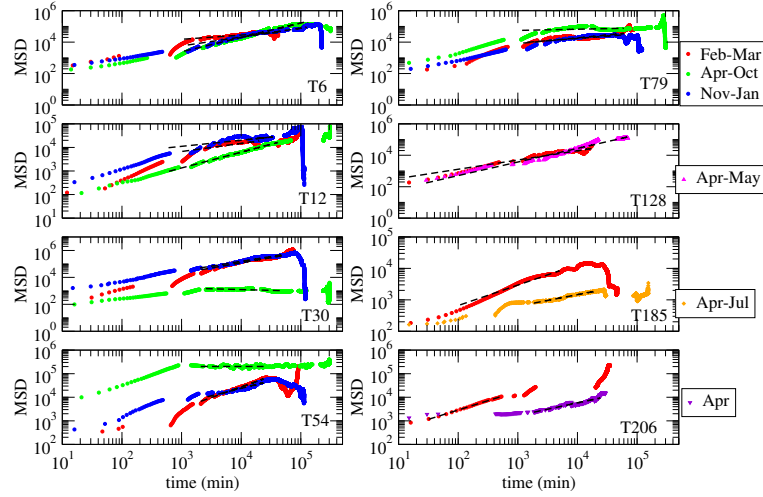


Figure 6: MSD segun separacion estacional para todas las tortugas.

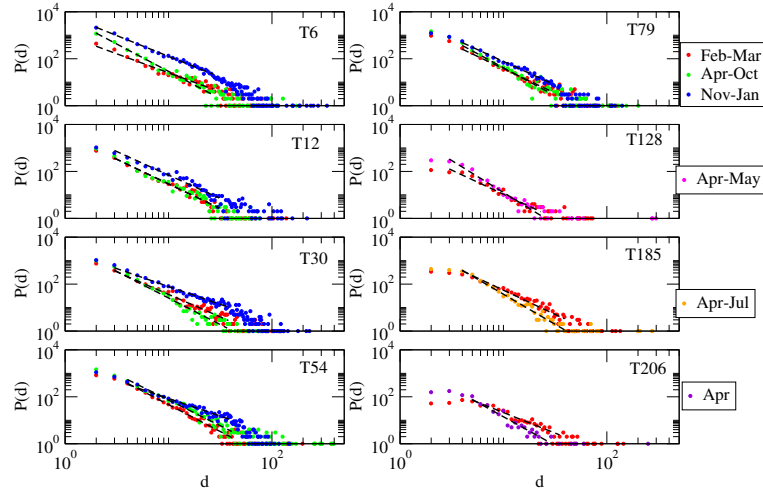


Figure 7: Distribucion de distancias recorridas en intervalos de 15min segun separacion estacional para todas las tortugas

Exponente alfa (MSD)				Exponente beta (P(d))			
Tortuga	Feb-Mar	Abril-Oct	Nov-Ene	Tortuga	Feb-Mar	Abril-Oct	Nov-Ene
T6	0.269	0.915	0.565	T6	1.62	2.3	1.76
T10	0.064	0.539	0.883	T10	1.8	2	2.4
T12	0.34	0.67	0.262	T12	2.1	2.1	1.99
T30	0.774	-0.117	0.708	T30	1.92	2.2	1.51
T54	0.901	0.001	0.629	T54	2.29	2.2	1.5
T79	0.364	0.075	0.292	T79	2.08	2.4	2.01
T128	0.588	0.865	S/D	T128	2.1	2.7	
T185	0.651	0.367	S/D	T185	2.1	2.5	
T206	0.792	0.54	S/D	T206	1.75	2.4	
Valor medio	0.5(3)	0.4(4)	0.5(2)	Valor medio	2.0(2)	2.3(2)	1.9(3)

Figure 8: Table 1: Exponents of MSD and $P(d)$ for each turtle and for each period. Males and females are distinguished by color, and the exponents were calculated using all available data.

Bouts of walking behaviour

We have shown that the power-law distribution of displacements can be explained by assuming the tortoise performs random walks with constant speed interspersed with resting periods, provided the duration of the walking bouts

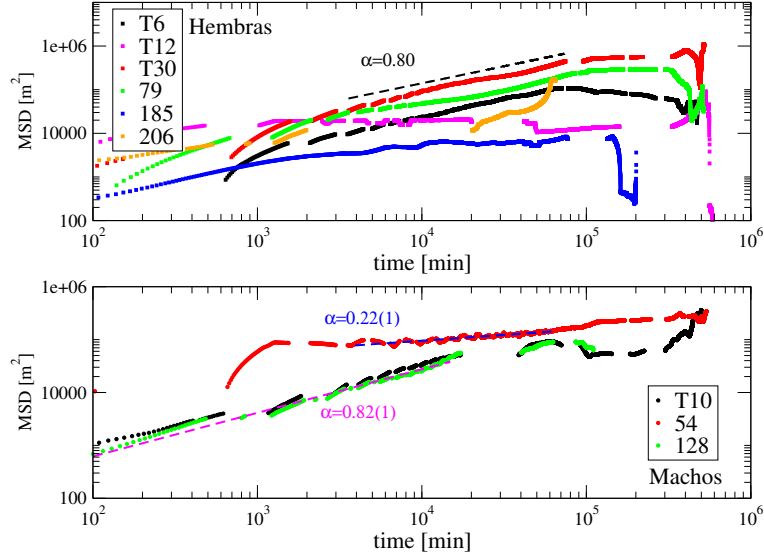


Figure 9: MSD separated in males and females

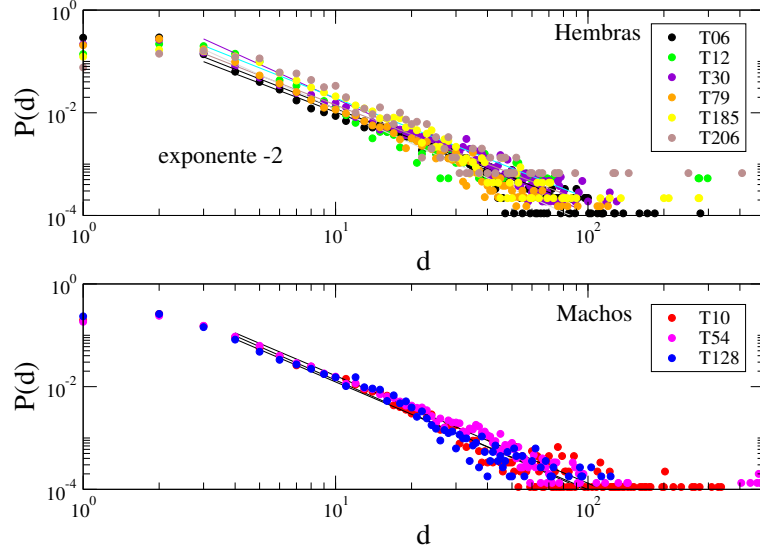


Figure 10: $P(d)$ vs d separated in males and females

is distributed with a power-law, or Pareto, distribution,

$$P_b(t) = \frac{\alpha - 1}{t^\alpha}, \quad t \geq 1,$$

where $P_b(t) dt$ is the probability that a bout of walking will last from time $t = 0$ to a time between t and $t + dt$.

This behaviour can also be explained by considering the transition probability from walking to resting, $\gamma(t)$. The transition probability is the probability that the tortoise will stop walking at a time between t and $t + dt$, given that it is walking at time t , i.e.

$$\gamma(t) = \frac{P_b(t)}{\text{P. of walking at time } t} = \frac{P_b(t)}{1 - \text{P. of having stopped between } 0 \text{ and } t} = \frac{P_b(t)}{1 - \int_0^t P_b(t') dt}.$$

Since $\int_1^t (\alpha - 1)t^{-\alpha} = 1 - t^{1-\alpha}$, we obtain

$$\gamma(t) = \frac{\alpha - 1}{t}, \quad t \geq 1,$$

		α (MSD)	β (P(d))
Hembras	T6	0.83	1.88
	T12		1.92
	T30	0.78	1.91
	T79	0.74	2.22
	T185	0.79	2.22
	T206	0.53	1.95
Machos	T10	0.82	2.16
	T54	0.22	2.09
	T128	0.82	2.09

Figure 11: Table 2: MSD and $P(d)$ exponents for each tortoise, separated by males and females, calculated using all available data without separation by periods.

i.e. the probability that the tortoise will decide to rest gets smaller the longer it has been walking.

Power-law-distributed bouts of behaviour have been discussed before [?], [?], but not everyone seems to agree on power laws [?].

Transition rate on a fixed maximum behaviour time

It has been shown that tortoises can be classified into two groups, in which both follow a power law distribution, $f(t)$, of exponent α with a maximum time of behaviour, t_{max} , [?]. In order to facilitate the theoretical analysis, we can infer a transition rate from one state to another, which is a function of the time spent in the current state, t . Another thing to consider is that in our simulation we have discrete time steps, δt . We will obtain the transition rate, through the following relation:

$$\gamma(t) = 1 - P(f(t + \delta t)|f(t)) \quad (3)$$

Where $P(f(t + \delta t)|f(t))$ is the conditional probability of being in the same state at time $t + \delta t$ given that it was in that state at time t . This conditional probability can be expressed as:

$$P(f(t + \delta t)|f(t)) = P(f(t) \cap f(t + \delta t)) \frac{P(f(t + \delta t))}{P(f(t))} \quad (4)$$

in which $P(f(t) \cap f(t + \delta t)) = 1$ and $f(t) = At^{-\alpha}$ with A a normalization constant. $P(f(t))$ can be calculated as follows:

$$P(f(t)) = \int_t^{t_{max}} f(t') dt' = \frac{A * (t_{max}^{1-\alpha} - t^{1-\alpha})}{1 - \alpha} \quad (5)$$

in this equation we are assuming that the state began at time $t = 0$ so if we apply this to the conditional probability we have:

$$P(f(t + \delta t)|f(t)) = \frac{t_{max}^{1-\alpha} - (t + \delta t)^{1-\alpha}}{t_{max}^{1-\alpha} - t^{1-\alpha}} \quad (6)$$

Finally, we can express the transition rate as:

$$\gamma(t) = \frac{(t + \delta t)^{1-\alpha} - t^{1-\alpha}}{t_{max}^{1-\alpha} - t^{1-\alpha}} \quad (7)$$

Supplement

By testing a different temporal interval, $\Delta t = 30$ min, the exponent does not change, nor is any improvement observed in $P(d)$ (Figure ??).

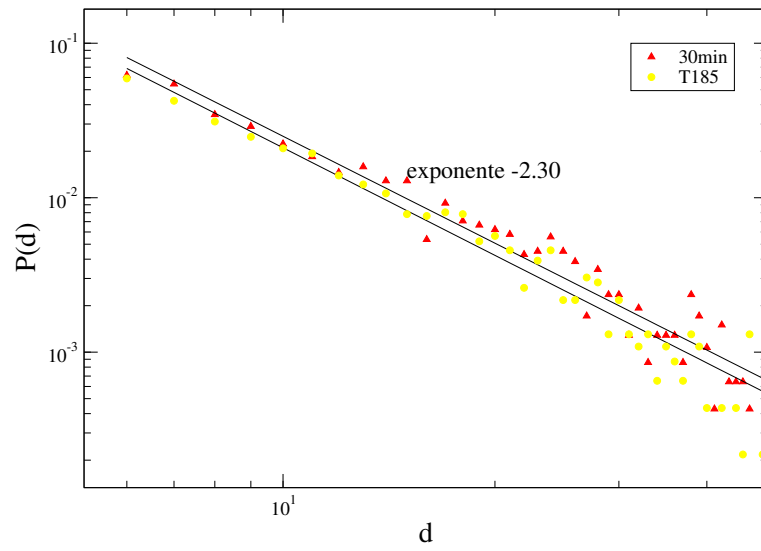


Figure 12: Distribution of traveled distances for T185, using $\Delta t = 15$ min and $\Delta t = 30$ min.